# Moment of Inertia: Verifying τ = Iα on a Spinning Disc

**SYNOPSIS**

Newton's second law has a rotational twin: τ = Iα. A torque applied to a rigid body gives it an angular acceleration, and the constant of proportionality is the body's moment of inertia — its resistance to being spun up, the rotational counterpart of mass. This lab puts that claim on a graph. A metal disc turns on a low-friction axle with a small pulley on its shaft; a string wound round that pulley runs over to a hanging mass, and the falling mass applies a torque. Change the hanging mass and you change the torque; measure the angular acceleration each time. If τ = Iα is right, those two quantities are proportional, and the slope of α against τ is 1/I. Then comes the real test: the disc is a uniform cylinder, so its moment of inertia can also be calculated from nothing but its mass and its radius. Two routes to the same number, one experimental and one geometric, and no reason they should agree unless the physics holds.

**Due** see the Module 5 Lab folder on Brightspace

**Report** Full lab report — use the Lab Template

**Pairs with** M5 — Moment of Inertia Lab Analysis

**Data** provided — this is a virtual lab

**PERFORMANCE EXPECTATIONS** what you should be able to do when this lab is written up

- **PE1** **Draw a free-body diagram for the hanging mass and a torque diagram for the disc**, and recognise that they are connected by one string carrying one tension.
- **PE2** **Explain why the tension is not equal to the weight of the hanging mass**, and derive $T = M_H(g - a)$ from $\Sigma F = M_H a$.
- **PE3** **Use the rolling constraint $a = r_p\alpha$** to tie the linear motion of the falling mass to the rotation of the disc, and say why the *pulley* radius is the one that appears and not the disc radius.
- **PE4** **Compute a torque as $\tau = r_p T$** for each run, carrying units, and recognise that τ is not simply proportional to $M_H$.
- **PE5** **Obtain a moment of inertia from the slope of α against τ**, with its uncertainty, and state why the slope is 1/I rather than I.
- **PE6** **Compare an experimental moment of inertia against the geometric value $\frac{1}{2}MR^2$** and judge the agreement against a stated criterion.

## This lab is Note 4, measured

**The analysis for this lab was worked out in class, by hand, before any data were taken.** The free-body diagram for the hanging mass, the torque diagram for the disc, the substitution that eliminates the tension, and the decision to plot α against τ rather than the other way up — all of that is the Moment of Inertia Lab Analysis note. Everything below is that argument written out. You are expected to bring the note into the report and build on it, not re-derive it from scratch as though you had never seen it.

Where each part of the note belongs in this report. The rubric awards points for these.

| Report section | What the note gives you |
|---|---|

| Report section | What the note gives you |
|---|---|
| **Introduction** | $\tau = I\alpha$ stated as the rotational second law, and **your own derivation** of the working relation — from the two diagrams, through $T = M_H(g - a)$ and $a = r_p\alpha$, to a torque written entirely in terms of things you can measure. End with a hypothesis you can confirm or reject. |
| **Experimental** | The side-on and top-down sketches of the apparatus, redrawn by you, with $r_p$ and $r_D$ marked and clearly distinguished. State the sign convention: which way is positive for the falling mass and which way for the rotation. |
| **Analysis** | The plot of $\alpha$ against $\tau$, with **slope = 1/I**, and the independent calculation $I_D = \frac{1}{2}M_D r_D^2$. |
| **Discussion** | Whether the two moments of inertia agree, which one you would trust, and what in the apparatus the model leaves out. |

## The physics you are using

Three relations, applied in order. The goal of the lab, stated plainly, is **to verify $\tau = I\alpha$**.

### One — the hanging mass

The falling mass has two forces on it: its weight down, and the string tension up. Taking down as positive,

$$\Sigma F = F_{net} = M_H a = F_{g,H} - T \quad \rightarrow \quad T = M_H g - M_H a = M_H(g - a)$$

This is the step people skip. **The tension is not the weight.** If it were, the mass would hang motionless. It is less than the weight by exactly the amount needed to accelerate the mass downward, and the faster the system accelerates the bigger that shortfall is.

### Two — the disc

The string leaves the pulley tangentially, so it applies a torque about the axle of magnitude $r_p T$, where $r_p$ is the *pulley* radius. The rotational second law gives

$$\tau = r_p T = I_D \alpha$$

### Three — the constraint that joins them

The string does not slip, so the speed of the string equals the speed of the pulley rim, and the linear acceleration of the hanging mass equals the tangential acceleration at the pulley:

$$a = r_p \alpha$$

Substituting that into the tension, and the tension into the torque, puts everything in terms of what you can actually measure:

$$r_p \left[ M_H(g - r_p\alpha) \right] = I_D \alpha$$

You **vary $M_H$** and you **measure $\alpha$**. Everything else in that line is a constant of the apparatus.

**Two radii, and they are not interchangeable.** The string winds on the small pulley on the shaft, radius $r_p$ = 1.45 cm. The spinning body is the disc, radius $r_D$ = 4.45 cm. **$r_p$ is the one in the torque and in the constraint; $r_D$ is the one in $\frac{1}{2}M_D r_D^2$.** They differ by a factor of three, so swapping them does not produce a slightly wrong answer — it produces one wrong by about an order of magnitude. Mark both on your sketch and label every use.

**Predict first. You double the hanging mass from 50 g to 100 g. What happens to the torque on the disc?**

**(A)** It exactly doubles — torque is $r_p M_H g$.

**(B)** It roughly doubles, but falls a little short, because the tension is less than the weight.

**(C)** It more than doubles, because the system also speeds up.

**(D)** It stays the same — the disc is unchanged.

**Why that question is the whole care of this lab.** It is tempting to write $\tau = r_p M_H g$, treating the hanging weight as the thing applying the torque. That is wrong, and it is wrong by a growing amount: the heavier the hanging mass, the faster the system accelerates, and the further the tension falls below the weight. In this data set the shortfall runs from about 5% at 50 g to about 9% at 100 g. Because the error grows with the independent variable, **it bends the graph rather than shifting it** — which is the kind of mistake a straight-line fit will happily absorb into a wrong slope without ever looking wrong. The correct torque is $r_p M_H (g - a)$, and your Analysis must show it.

## The apparatus

A metal disc is mounted on a low-friction axle so it can spin freely in a horizontal plane. A small pulley is fixed to the shaft at the centre of the disc. A string is wound round that pulley, run horizontally to the edge of the bench, over a guide, and down to a hanger. Releasing the hanger lets it fall; the string unwinds and spins the disc. A sensor records the angular acceleration of the disc for each run.

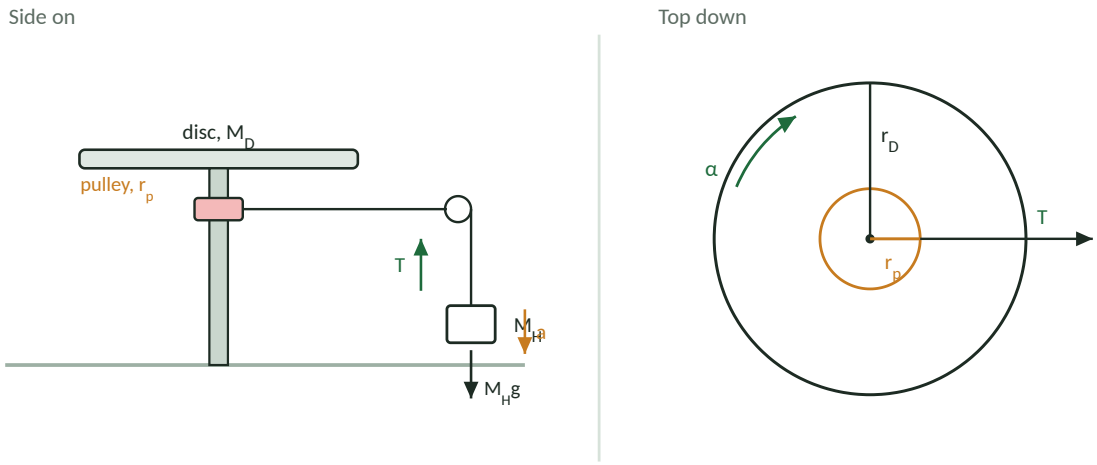



Figure 1. The apparatus. Left, side on: the falling mass unwinds the string from the shaft pulley. Right, top down: the string leaves the pulley tangentially, so the torque arm is $r_p$, while the body being spun has radius $r_D$.

### The constants

| Quantity | Measured | In SI |
|---|---|---|
| Pulley diameter | 2.90 cm | $r_p = 0.01450$ m |
| Disc diameter | 8.90 cm | $r_D = 0.04450$ m |
| Disc mass | 212.8 g | $M_D = 0.2128$ kg |

From those three numbers alone, the geometric moment of inertia of a uniform disc about its centre is

$$I_D = \tfrac{1}{2} M_D r_D^2 = \tfrac{1}{2}(0.2128)(0.04450)^2 = 2.107 \times 10^{-4}\ \text{kg·m}^2$$

That is the number your graph has to reproduce. **It was calculated before the experiment, not fitted to it**, which is what makes the comparison a test rather than a circular argument.

## Part I — The data

Six runs. The hanging mass was increased in 10 g steps and the angular acceleration of the disc recorded for each.

The six runs as recorded. Everything else in this lab is derived from these two columns.

| Run | Hanging mass $M_H$ (g) | Angular acceleration α (rad/s²) |
|---|---|---|
| 1 | 50 | 33 |
| 2 | 60 | 40 |
| 3 | 70 | 46 |
| 4 | 80 | 50 |
| 5 | 90 | 57 |
| 6 | 100 | 63 |

## Part II — Turning each run into a torque

Four steps per run, in this order. Each is a sample calculation the rubric asks to see worked once.

I. **The linear acceleration.** $a = r_p\alpha$. Note that this is *not* a measured quantity — it is the measured α converted through the constraint.

II. **The tension.** $T = M_H(g - a)$, with $g = 9.8$ m/s². Compute $M_Hg$ and $M_Ha$ separately the first time so you can see the size of the correction.

III. **The torque.** $\tau = r_pT$. Units are N·m; they are not joules, and saying so is worth a sentence.

IV. **Tabulate.** $M_H$, α, a, $M_Hg$, $M_Ha$, T, τ, one row per run.

> **Look at the tension column before you go on.** The weight $M_Hg$ rises in even 0.098 N steps because the masses do. The tension does not: it rises more slowly, and the gap between them widens run by run. That widening gap is the whole content of step 2, and a reader of your table should be able to see it without being told.

## Part III — The graph that measures the moment of inertia

From $\tau = I\alpha$, solved for the quantity you measured:

$$\alpha = ( 1 / I ) \times \tau$$

So plot **α up the page against τ across it**, and the slope is **1/I** — not I. Invert it to get the experimental moment of inertia:

$$I_{expt} = 1 / \text{slope} \qquad \delta I = \delta(\text{slope}) / (\text{slope})^2$$

### The comparison

Put the two moments of inertia side by side: $I_{expt}$ from the slope, and $I_D = \tfrac{1}{2}M_D r_D^2 = 2.107 \times 10^{-4}$ kg·m² from the geometry. Quote a percent difference. A difference of **10% or less** is a confirmation of $\tau = I\alpha$.

If you land outside that, check in this order: $r_D$ used where $r_p$ belongs; the slope reported as I instead of 1/I; the torque built from $M_H g$ instead of $M_H(g - a)$; or diameters used in place of radii.

## Check your analysis

Paste your six torque-and-angular-acceleration pairs to check the line you fitted. Your report still has to show the graph and a worked sample calculation of each type, by hand, with units.

**TOOL · FIT A AGAINST T, AND READ OFF THE MOMENT OF INERTIA**

**One run per line: τ in N·m, then α in rad/s², separated by a comma, a tab or a space**

**Disc mass (kg)**

0.2128

**Disc radius (m)**

0.0445

The slope is 1/I, so the uncertainty on I is δ(slope)/(slope)² — the tool applies that for you, but AQ4 asks you to derive it.

## Additional Questions

**Choose three of the five below and answer those three fully.** Name which three you chose. Answer them in a section of their own headed **Additional Questions**, in **sentences and paragraphs** with the working shown, never as a bare numbered list and never by restating the question before answering it. Depth on three beats coverage of five; this section is worth 8 points on the rubric.

**AQ1 — how wrong is the easy mistake?** Redo the whole analysis using the incorrect torque $\tau = r_p M_H g$, as though the tension equalled the weight. Report the slope, the moment of inertia it gives, and its percent difference from $\tfrac{1}{2}M_D r_D^2$. Then explain the *shape* of the error: show that the fractional overestimate of the torque is a/g, that a grows with $M_H$, and that the consequence is a graph which is still convincingly straight but has the wrong slope. Say what this implies about using $R^2$ as evidence that an analysis is correct.

**AQ2 — where the string's own mass went.** The model treats the string as massless and the guide pulley as frictionless and inertia-free. Pick *one* of those idealisations and work out which way relaxing it would move your experimental moment of inertia — higher or lower than the true disc value. Justify the direction physically rather than by hand-waving. Then look at the sign of your own discrepancy and say whether it is consistent with the mechanism you chose. If it is not, say so plainly; a result that fails to support your explanation is still a result.

**AQ3 — the disc is not quite a uniform cylinder.** ½MR² assumes all the mass is spread evenly over a flat disc. The real object has a shaft and a pulley attached at its centre. Using the parallel-axis theorem where appropriate and the formula for a solid cylinder, estimate the moment of inertia contributed by the pulley (treat it as a small uniform cylinder of radius $r_p$ and guess a plausible mass). Express it as a percentage of the disc's own moment of inertia. Does including it move the geometric value toward or away from your experimental one, and is the size of the correction big enough to matter at the precision you achieved?

**AQ4 — propagating an inverted slope.** You measured a slope $s = 1/I$ and reported $I = 1/s$. Derive $\delta I = \delta s/s^2$ by differentiating $I = s^{-1}$, and show that the *fractional* uncertainty is unchanged by the inversion: $\delta I/I = \delta s/s$. Apply it to your own fit and quote $I_{expt} \pm \delta I$. Then answer the question that matters: is the difference between your experimental and geometric moments of inertia larger or smaller than $\delta I$? If it is smaller, what exactly have you demonstrated — and what have you not?

**AQ5 — predicting a run you were not given.** Using your experimental moment of inertia and the working relation $r_p[M_H(g - r_p\alpha)] = I\alpha$, solve algebraically for $\alpha$ in terms of $M_H$ and the constants — note that $\alpha$ appears on both sides, so this takes a line of rearrangement. Then predict the angular acceleration for a hanging mass of 150 g, and for 500 g. Comment on what your expression says happens as $M_H$ becomes very large: does $\alpha$ grow without limit, and if not, what does it approach and why is that physically sensible?

## Prompts for your Discussion

These are **not** the Additional Questions. Work them into your **Discussion** section as part of your argument, in your own order. Do not answer them as a list, and do not limit yourself to them.

I. You have two moments of inertia for the same disc. Which would you report if you had to pick one, and what would have to be true for the experimental value to be the better of the two?

II. Your fitted line has an intercept. The model says $\alpha$ should be zero when $\tau$ is zero. Is your intercept large enough to be meaningful, and if so what would produce it?

III. Moment of inertia is often described as "rotational mass". Using your own numbers, say what is right about that description and what is misleading — in particular, why the same disc would have a different moment of inertia about a different axis while its mass would not change.

IV. The tension correction mattered at the few-percent level here. Describe a version of this experiment in which it would matter hardly at all, and one in which it would dominate.

V. The angular accelerations were recorded to two significant figures. Propagate that precision through to your moment of inertia and say whether it limits the comparison you made.

VI. If the string slipped on the pulley during a run, what would you expect to see in the data, and would it be distinguishable from a genuinely smaller moment of inertia?

## Files for this lab

HANDOUT

### This lab, as a PDF

The whole page, printable. The calculator only works on the web version.

REQUIRED

### Example workbook

The six runs, a blank analysis sheet laid out for you, and the whole set worked.

GRADING

### Lab 07 rubric

Read it before you write, not after.

SUPPORT

### Report starter

Section guidance only — the sentence stems have gone. See the note inside.

NOTE

**M5 — Moment of Inertia Lab Analysis**

The in-class derivation this lab is built on. Not yet published as a page; use your printed copy.

REFERENCE

**Formatting data tables**

Table captions, units in headers, significant figures.

## References to get you started

Your references should include material on rotational dynamics and moment of inertia. Your textbook, this handout and the Module 5 notes all count, and you need at least one source beyond the textbook.

1. [Author's names], *Journal or Book title*, edition, Volume number, page number, (year).

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